

PEDRAM AMINHARATI

AI & Control Engineer — Robotics and Manipulation

PhD in Mechatronics & Artificial Intelligence · Manipulation robotics on real hardware · Imitation learning & diffusion policies · ROS2, MuJoCo, Isaac Sim

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PROFILE

PhD in mechatronics and artificial intelligence (Arts et Métiers ParisTech – CNRS), working at the interface between model-based approaches and machine learning. Around five years of robotics: adapted a KUKA robot for automated fiber placement with AI-driven real-time trajectory correction, projects on Franka Emika Panda arms, control of a mobile manipulator, motion planning and trajectory generation.

Trained diffusion policies in simulation in MuJoCo, Isaac Sim and PyBullet, backed by deep practice in building and maintaining sensor datasets (synchronized multimodal acquisition, timestamping, augmentation). Expert in Python, C++ at read-and-modify level, ROS/ROS2. Two international journal publications, including a review article. Teaches robotics, industrial AI and digital twin at Arts et Métiers.

PROFESSIONAL EXPERIENCE

Research Engineer – Doctoral Researcher · Robotics, AI & Control

AMVALOR (Arts et Métiers technology transfer subsidiary) — PIMM (CNRS UMR 8006) & LCFC Laboratories, Paris | **Since Nov. 2023**

- **Adapted a KUKA robot for automated fiber placement (AFP) with AI-driven real-time trajectory correction:** online adjustment of the lay-up path and process parameters from sensor feedback, to hold deposition quality during production.
- **Manipulation robotics on real hardware:** projects on **Franka Emika Panda** arms and control of a **mobile manipulator** — motion planning, trajectory generation and execution, kinematics and dynamics.
- **Trained diffusion policies in simulation** in MuJoCo, Isaac Sim and PyBullet; simulator comparison and assessment of physical realism.
- **Adapted object detection models** (vision) for identifying parts in an industrial context.
- **Built and maintained sensor datasets:** real-time synchronized multimodal acquisition architecture, stream timestamping, cleaning, correction and augmentation of data for learning.
- **Hybrid physics/data modelling** (physics-informed neural networks): numerical optimisation, linear algebra and experimental validation on an instrumented test bench I designed and built myself.
- **Designed, tested and improved detection, identification and severity-grading algorithms** from sensor signals: signal processing, feature extraction, machine learning and deep learning.
- **Technology watch and literature review:** authored a review article published in an international journal; regularly presented methods and results to research teams and industrial partners.
- **Teaching at Arts et Métiers: Robotics** and "Industrial Applications of AI & Digital Twin" — integrating, managing and validating AI components inside digital twin workflows, from simulation to industrial deployment.

Mechatronics and Robotics Laboratory Manager

K. N. Toosi University of Technology — Tehran, Iran | **Nov. 2020 – Aug. 2022**

- **Led the laboratory's robotics activities** and managed its experimental equipment.
- **Developed FDM 3D printers** achieving **25 µm** accuracy: mechanical design, axis motion control, sensor instrumentation and experimental tuning.
- **Designed and developed smart pill dispensers** for dementia patients: complete mechatronic system from mechanics to electronics and control.

Teaching Assistant — Robotics and Mechatronics

K. N. Toosi University of Technology — Tehran, Iran | **Oct. 2019 – Aug. 2022**

- Tutorials and lab sessions for **Robotics** and **Introduction to Mechatronics**.

Apprentice — Advanced Industrial Automation Technology

Festo — Tehran, Iran | **Oct. 2020 – Mar. 2021**

- **Troubleshoot and debugged pneumatic and electric control systems:** sensor feedback, actuator response and PLC programming.

R&D Manager (alongside engineering studies)

- R&D on vehicle parts assembly: dimensional specifications, blueprint interpretation and quality control.

EDUCATION

PhD in Mechatronics and Artificial Intelligence

Arts et Métiers ParisTech (ENSAM Paris) – CNRS | Nov. 2023 – Jan. 2026

Thesis: hybrid approach coupling physics-based modelling and artificial intelligence, experimentally validated, for analysing the behaviour of composite structures under extreme conditions.

Master's — Artificial Intelligence & Cybersecurity

Universität Klagenfurt, Austria | Oct. 2022 – Aug. 2024

Supervised, unsupervised and reinforcement learning; deep neural networks; computer vision.

M.Sc. — Mechatronics, Robotics and Automation Engineering

K. N. Toosi University of Technology, Tehran | Sept. 2019 – 2022

Thesis: optimisation of 3D printing process parameters using an artificial neural network model.

B.Sc. — Mechatronics, Robotics and Automation Engineering

Sharif University of Technology, Tehran | 2013 – 2018

PUBLICATIONS

- **Aminharati P.**, Shirinbayan M., Fitoussi J., Benfriha K., Meraghni F. — *AI-driven advances in composite materials for hydrogen storage vessels: A review*. International Journal of Hydrogen Energy, Vol. 171 (2025), 151288.
- Nikooharf M. H., Shirinbayan M., Ghodsian N., **Aminharati P.**, Bahlouli N., Fitoussi J., Benfriha K. — *Toward advance/digitalized FFF: real-time multimodal synchronized data acquisition and ML/DL-driven process optimization*. Progress in Additive Manufacturing, 10 (2025), 9195–9212.

TECHNICAL SKILLS

Robotics & manipulation KUKA (automated fiber placement, real-time trajectory correction) · Franka Emika Panda · Mobile manipulator · Motion planning and trajectory generation · Kinematics and dynamics · ROS / ROS2 · Real robot data analysis

Simulation MuJoCo · Isaac Sim · PyBullet · Gazebo · CoppeliaSim · Simulator comparison and physical realism · Digital twin

AI & learning Diffusion policies · Imitation learning · Reinforcement learning · Physics-Informed Neural Networks · Machine learning and deep learning · Dataset construction, correction and augmentation

Programming Python (expert: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Jupyter) · C++ (read and modify) · MATLAB · Git · Software best practices and automated testing

Control & computation Control theory, observers and controllers · Numerical optimisation · Linear algebra and matrix computation

Vision Object detection · Feature extraction · Signal processing and vibration analysis

Instrumentation & testing Test bench design and build · Synchronized multimodal acquisition and timestamping · Sensors (vibration, pressure, position, temperature) · PLC programming (Festo)

Mechanical design SolidWorks · CATIA V5 · AutoCAD · Abaqus (finite element)

LANGUAGES

English Fluent (C1) · **Persian** Native · **French** Basic (A1), currently learning